

CariSurg MedTech Pathways Research Project

Preliminary Proposal Template

Project Title

Development and Evaluation of an AI-assisted Emergency Department Triage Model for Improved Clinical Decision-Making

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Statement of the Problem

Emergency department clinicians must prioritize patients although there is overcrowding and large patient volumes, which leads to delays in critical care and inconsistent triage decisions in the hospital emergency departments (1,1). Machine learning models have demonstrated an improved performance but many have been developed using a single dataset. Therefore, they have not been extensively evaluated across diverse healthcare settings including the Caribbean populations (2). This project proposes developing and evaluating an AI-assisted triage model to improve triage consistency that can help clinicians interpret the model to support clinical decisions.

Previous Work

The research has demonstrated that artificial intelligence can improve the accuracy, precision and timing of the emergency department triage. (1)found that AI and machine learning models generally outperformed conventional triage methods in predicting and identifying critically ill patients which improves resource allocation and support for patients in need of critical care.

(2)developed and compared seven machine learning models using the Korean Triage and Acuity Scale (KTAS). Their stacking and ensemble model achieved an overall performance of 80.05% accuracy which only relied on a single dataset that the authors acknowledged is a limitation of generalizability.

(3)addresses the issue by performing external validation of a machine learning triage model across multiple hospitals using a common data model. Their findings demonstrated strong predictive performance that emphasizes the importance of testing AI systems beyond a single institution before deploying to other institutions.

(4)proposes an interpretable machine learning triaging model that addresses class imbalance that maintains transparency. The study demonstrated that interpretable models can achieve strong predictive performance, then it highlights that there is a growing importance of clinician trust in AI-assisted healthcare systems.

(5)developed an explainable machine learning model for rapid risk stratification in emergency departments, demonstrating that transparent AI systems can provide an accurate prediction that

allow clinicians to understand the factors that influence model outputs and increase confidence in AI supported healthcare

(6) reviewed the benefits and challenges of AI driven triaging systems and concluded that issues such as clinician trust, algorithmic bias, and ethical considerations remain a significant barrier to widespread implementation despite its promising predictive performance.

Together these studies demonstrate that AI can improve in triaging in the emergency department. It also reveals important gaps related to validation and testing in different diverse populations and the need for models that are highly interpretable to clinicians.

(7) developed machine learning and natural language processing models to predict patient disposition using emergency department triage information. The study showed that incorporating both structured triage variables and clinical text improved predictive performance for patient outcomes. However, the models were developed using retrospective data which require further validation in diverse healthcare environments before implementing them in different hospitals. Their further findings support the need for AI systems that can generalize across different populations and clinical settings.

Proposed Solution

This project proposes that the AI-assisted emergency department triaging model should use machine learning techniques trained on emergency department data. The model will be used to predict patient triage levels with emphasizing both predictive accuracy and ensuring that it is interpretable by clinicians. The project will address the research gaps by evaluating the model's applicability to the healthcare settings in different demographic groups including the Caribbean and by incorporating machine learning techniques that allow clinicians to understand the factors influencing model predictions. The goal is to support a more consistent triage decision and improve clinical decision making in a resource constrained emergency department.

References

- 1. Tyler S, Olis M, Aust N, Patel L, Simon L, Triantafyllidis C, et al. Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. Cureus. 2024 May 8. doi:10.7759/cureus.59906**
- 2. Araouchi Z, Adda M. TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. Procedia Comput Sci. 2024;251:430–7. doi:10.1016/j.procs.2024.11.130**
- 3. Yu JY, Kim D, Yoon S, Kim T, Heo S, Chang H, et al. Inter hospital external validation of interpretable machine learning based triage score for the emergency department using common data model. Sci Rep. 2024 Mar 20;14(1):6666. doi:10.1038/s41598-024-54364-7**
- 4. Look CS, Teixayavong S, Djärv T, Ho AF, Tan KB, Ong ME. Improved interpretable machine learning emergency department triage tool addressing class imbalance. Digit Health. 2024 Jan;10:20552076241240910. doi:10.1177/20552076241240910**

5. Van Doorn WPTM, Helmich F, Van Dam PMEL, Jacobs LHJ, Stassen PM, Bekers O, et al. Explainable Machine Learning Models for Rapid Risk Stratification in the Emergency Department: A Multicenter Study. *J Appl Lab Med*. 2024 Mar 1;9(2):212–22. doi:10.1093/jalm/jfad094
6. Da'Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *Int J Med Inf*. 2025 May;197:105838. doi:10.1016/j.ijmedinf.2025.105838
7. Chang YH, Lin YC, Huang FW, Chen DM, Chung YT, Chen WK, et al. Using machine learning and natural language processing in triage for prediction of clinical disposition in the emergency department. *BMC Emerg Med*. 2024 Dec 18;24(1):237. doi:10.1186/s12873-024-01152-1